

Highlights

Early fault detection and estimation in wine fermentation based on the extended Kalman filter

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- A **dynamic model** of the wine fermentation process is used for monitoring.
- EKF was applied to detect stuck and sluggish wine fermentations early.
- The performance of the proposed method was analyzed using Monte Carlo scenarios.
- Meaningful sugar measurements are required for accurate prediction.
- A reliable detection is possible on the first day by measuring the initial nitrogen.

Early fault detection and estimation in wine fermentation based on the extended Kalman filter

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Abstract

This study presents a state estimation strategy using an Extended Kalman Filter (EKF) integrated with **dynamic models** for the early detection of stuck and sluggish wine fermentations. The proposed approach aims to provide enologists with accurate, real-time information throughout fermentation, enabling timely corrective actions and informed decision-making. The analysis investigates the sensitivity of the estimation performance to key process factors, including model parameters and initial conditions, with particular emphasis on the role of unmeasured nitrogen levels. The results highlight the critical dependence of accurate state estimation on sugar concentration data, which are typically unavailable during the **exponential** phase. Monte Carlo simulations demonstrate that the proposed algorithm can identify 95% of problematic fermentations within the initial days of the process. Furthermore, incorporating initial nitrogen measurements **reduces the time of the first alarm by at least 65%, justifying additional cost**. These findings underscore the potential of EKF-based **estimation** frameworks to enhance monitoring and control strategies in wine production and other alcoholic fermentation processes.

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1. Introduction

Alcoholic fermentation, which involves the conversion of grape sugars into ethanol and carbon dioxide by yeasts, is a crucial step in winemaking. This complex microbial process requires optimal conditions to complete successfully; otherwise, it can lead to problematic fermentations with significant economic losses. Factors that affect fermentation include toxicity due to high ethanol, low pH, extreme temperatures, and the limitation of key nutrients such as nitrogen and oxygen (Bisson, 1999).

Common problems encountered in industrial wineries include *stuck* fermentations, characterized by a high level of residual sugar remaining at the end, and *sluggish* fermentations, marked by a low rate of sugar consumption on the last days (Malherbe et al., 2007). These problems result in reduced productivity, lower wine quality, and, in the worst case, the loss of raw materials. Early detection of potential faults is essential to enable timely preventive interventions. This challenge in winemaking underscores the need for reliable techniques to identify problematic fermentations at an early stage and to evaluate the most effective corrective strategies.

In the industry, the enologist usually rely on frequent density measurements to detect an eventual fault and prevents it by acting on nutrition and temperature. To improve this process, online monitoring is seen as a promising method, as it can support the enologist in decision-making during fermentation and allows the adoption of novel control techniques (Sablayrolles, 2009). One of the requirements for this is the implementation of sensors to measure variables related to fermentation rate such as density or CO₂ production. This information can then be used to constantly update accurate models of the process, enabling the estimation of unmeasurable variables and the prediction of future behavior. If the model provides an accurate description of the physical process, it could be regarded as a virtual copy of process, or a *digital twin* (Jones et al., 2020).

27 Digital twins combine physical entities with virtual models connected through con-
28 tinuous data exchange, enabling real-time monitoring, simulation, and control. The
29 representation of the physical process provide the foundation for these virtual models
30 allowing accurate prediction and optimization. The approach for building the digital
31 twin can be classified into two main categories: *data-driven* and *model-driven* meth-
32 ods (Kadlec et al., 2009). Data-driven methods rely on large datasets to build black-box
33 models, while model-driven methods use first-principles equations to describe the un-
34 derlying physical and chemical phenomena (Jiang et al., 2021). For both methods,
35 state estimation, e.g. the Extended Kalman Filter (EKF), are essential to keep the dig-
36 ital twin synchronized with the process since it enables a real-time update of states
37 and parameters of the model. ~~This integration of model-based computation is what~~
38 ~~sets digital twins apart from traditional simulations, making them powerful tools for~~
39 ~~process optimization and decision support.~~

40 There are several data-driven methods for monitoring in the process industry such
41 as Principal Component Analysis (PCA), Neuro-Fuzzy Systems, and Artificial Neu-
42 ral Networks (ANN) (Kadlec et al., 2009). This class of methods requires a large
43 amount of data for building black-box predictive models, but limited knowledge about
44 the underlying process is needed. In contrast, model-driven methods are based on first
45 principles equations describing the chemical and physical phenomena of the process
46 (Jiang et al., 2021). Although these methods usually assume several simplifying con-
47 ditions, they are useful for describing the internal variables of the process that are not
48 directly measured. These two approaches can also be used together in the development
49 of hybrid models that could result in better prediction and extrapolation using the same
50 amount of training data (von Stosch et al., 2014). For instance, Luna et al. (2023) de-
51 veloped a hybrid model that integrates a mechanistic model for alcoholic fermentation
52 with a machine learning algorithm for predicting the extraction of polyphenols in red
53 wines. Another example is Coleman et al. (2007) which models the wine fermentation
54 with Michaelis-Menten style kinetics, but uses polynomials for the reaction rate terms
55 to fit the data.

56 Data-driven approaches for monitoring and fault detection of wine fermentation
57 can be found in the literature. Urtubia et al. (2007) employed PCA and cluster analy-

sis (K-means) of data from 24 industrial Cabernet Sauvignon fermentations to detect anomalous behaviors in advance, using data on sugar, alcohol and organic acids. In this case study, it was possible to predict the likelihood of the process ending normally using only data from the first three days of fermentation. In later studies, the authors compared the prediction provided by multiway PCA, multiway partial least squares, and ANN on the same data, achieving 100 % accuracy (Urtubia et al., 2012). Despite achieving good results, the methods rely on measurements that are difficult to acquire in an industrial setting, such as alcohol and organic acids, limiting their applicability.

Among model-driven methods, nonlinear observers and filters, such as the Extended Kalman Filter (EKF), are among the most popular (Lyubenova et al., 2021). In Bárzaga-Martell et al. (2021), the authors combine an EKF with an asymptotic observer for the estimation of biomass and ethanol from high-density yeast cultures using measurements of dissolved oxygen, CO₂ and glucose. The proposed observer was able to accurately predict the switch between different metabolic regimes, which is an important sign of overflow metabolism. Sipos (2020) develop an EKF for wine fermentation considering measurements of temperature and CO₂ to estimate biomass and sugar concentrations. However, different models were used for the decay and exponential/log phases, increasing the complexity of the implementation.

Kalman filtering is also used for *fault detection*, where the fault is incorporated into the model equation and estimated along with the states (Kordestani et al., 2021). For example, the process can be monitored for operational problems or component degradation. When coupled with some kind of uncertainty propagation, the EKF can also be used for *fault prognosis*, where the future status of the process is predicted (Akkermans et al., 2018). In the Extended Kalman Filter (EKF), the propagation of the mean and covariance relies on a first-order approximation of the system dynamics. While this may yield acceptable estimates for the mean in mildly nonlinear systems, the prediction of the covariance can be highly unreliable, particularly in the presence of strong nonlinearities, leading to inaccurate uncertainty quantification (Landgraf et al., 2023). Nevertheless, it has a low computational load and its implementation is simple compared to other methods such as Monte Carlo simulations and polynomial chaos expansion.

89 In this context, we propose a monitoring strategy for wine fermentation with a
90 model-based framework comprising an EKF for real-time state estimation and a batch-
91 end sugar level predictor for early fault detection. Unlike previous studies, the proposed
92 method combines the advantages of EKF with a first-principles model, to obtain a
93 robust algorithm for fault prediction and monitoring. Furthermore, the method could
94 be easily used in the industry since it only requires online measurements of density and
95 temperature, both already available on most wineries. The goal is to provide accurate
96 information in real time for the winemaker to support his decision-making in order to
97 reduce the likelihood of a bad batch occurring.

98 The proposed method, alongside the simulation studies, result in the following con-
99 tributions:

- 100 • A dynamic model of the wine fermentation process is developed based on a first-
101 principles model with EKF for online estimation of the states and parameters.
- 102 • Applying EKF for early detection of problematic wine fermentations through
103 final sugar prediction;
- 104 • Accurate prediction available online relying only on temperature and density
105 measurements;
- 106 • Analysis of the main measurements affecting the prediction of the final sugar
107 concentration;
- 108 • Analysis of the performance and reliability of the proposed method based on
109 Monte Carlo scenarios.

110 This document is organized as follows: in Section 2, the fermentation process and
111 EKF for fault detection are explained in details; in Section 3.1 a case study on white
112 wine fermentation is presented, followed by Monte Carlo scenarios on white and red
113 wine in Section 3.2; in Section 3.2.1 the effects of measuring initial nitrogen is ana-
114 lyzed; finally, the results are discussed in sections 3.3 and 4.

115 2. Methods

116 After some notation, the wine fermentation process and the dynamic model consid-
117 ered in this work are discussed in detail. Then, the proposed state estimation and fault
118 detection algorithms are presented.

119 2.1. Preliminaries and Notation

120 A continuous dynamic nonlinear model can be represented by:

$$\begin{aligned}\dot{x} &= f_c(x(t), u(t), p) \\ y(t) &= h(x(t))\end{aligned}$$

121 which has dimensions $x \in \mathbb{R}^{n_x}$ for the states, $u \in \mathbb{R}^{n_u}$ for the control inputs, $p \in \mathbb{R}^{n_p}$ for
122 the parameters, and $y \in \mathbb{R}^{n_y}$ for the measured outputs. $f_c(\cdot)$ and $h(\cdot)$ represent nonlinear
123 functions for the states and outputs, respectively. This model can be transformed to
124 discrete time by an integration method or by an ODE solver, resulting in:

$$\begin{aligned}x(k+1) &= f(x(k), u(k), p) \\ y(k) &= h(x(k))\end{aligned}\tag{1}$$

125 where $t = t_s k$ with t_s as the sampling time and k as the discrete sample. This notation
126 for discrete-time variables can be abbreviated to $x_k = x(k)$.

127 2.2. Wine Fermentation

128 The fermentation starts when the tank is inoculated with yeast, usually *Saccha-*
129 *romyces cerevisiae*, and ends when there is no sugar left, i.e. the wine is dry. This takes
130 around 6–10 days operating around 25 °C for red wines and 12–20 days at 15 °C for
131 white wines (Boulton et al., 1999). A diagram of a typical automated fermentation tank
132 can be seen in Figure 1, which shows the temperature control using service water and
133 sensors for temperature and density. Specifically for red wines, the juice is fermented
134 in contact with the solids in a process called fermentative maceration which finishes
135 when the must is pressed and the wine is separated (draining point). This process can
136 continue long after the wine is dry, i.e. extended maceration, depending on the style of
137 wine (Casassa and Harbertson, 2014).

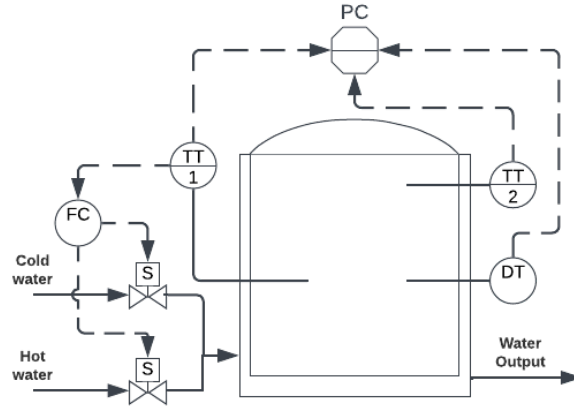


Figure 1: P&ID of the tank fermentation. TT1 and TT2 are temperature sensors for different sections of the tank. DT is a density sensor.

The management of nitrogen additions at the beginning and during the fermentation process is key since it is the major limiting nutrient for yeast growth (Santamaria et al., 2020). A suitable dosage during the process helps to synthesize secondary metabolites such as esters that contribute to the aromatic wine quality (Seguinot et al., 2018). Furthermore, the amount of nitrogen in the must is an important factor on fermentation kinetics: musts with higher nitrogen content lead to faster fermentations, whereas lower concentration results in slower fermentations. Inadequate levels of nitrogen can cause stuck or sluggish fermentations.

There are several sources of nitrogen available in grapes and musts, however, only some of them are easily assimilable by yeast such as ammonium and amino acids. Yeast Assimilable Nitrogen (YAN) is important because it is responsible for cell growth and metabolism, although it is expensive to measure accurately during fermentation (Bell and Henschke, 2005). In addition, there may be significant variability in YAN in vineyards, increasing the uncertainty of the process (Malherbe et al., 2007).

To ensure the success of wine fermentation, most industrial wineries have invested in automation technologies such as temperature control and automatic pump-overs, among others. In the last decade, affordable and reliable density sensors have been developed, which allow online monitoring of the fermentation progress (Sablayrolles,

2019). Density measurements based on differential pressure or production of CO₂ can be used to infer the sugar content in the fermenting must (El Haloui et al., 1988). In this work, it is assumed that sugar (S) and density (ρ) are related by the following linear equation:

$$\rho = \frac{100}{250}S + 1000 \quad (2)$$

with sugar in g/L and density in kg/m^3 . Furthermore, temperature (T) is assumed constant and known.

2.3. Fermentation Model

Most wine fermentation models are described by a set of nonlinear ODEs with specific assumptions and purposes, as reviewed in Miller and Block (2020). In Coleman et al. (2007), a simple lumped parameter model, assuming nitrogen as the primary limiting substrate and temperature-dependent kinetic parameters, was developed. This model is the one used in this work, with a few adjustments.

The continuous model equations are described by:

$$\dot{X} = \mu X - k_d E X, \quad \mu = \mu_{max} \frac{N}{K_N + N}, \quad (3a)$$

$$\dot{N} = -\mu \frac{X}{Y_{X/N}}, \quad (3b)$$

$$\dot{E} = \beta X, \quad \beta = \beta_{max} \frac{S}{K_S + S}, \quad (3c)$$

$$\dot{S} = -\beta \frac{X}{Y_{E/S}}, \quad (3d)$$

where the state variables are given by the biomass (X), nitrogen (N), ethanol (E) and sugar (S), while the other terms are given by the maximum specific growth rate (μ_{max}), maximum specific rate of sugar utilization (β_{max}), half-velocity constant for nitrogen (K_N), half-velocity constant for sugar (K_S), death rate by ethanol (k_d), the yield coefficient for cell mass grown per mass of nitrogen ($Y_{X/N}$) and yield coefficient for ethanol produced per sugar consumed ($Y_{E/S}$). These terms related to the kinetics, summarized in Table 1, are modeled as exponentials of polynomials up to order 2 dependent on temperature T or initial nitrogen N_0 .

The nominal values for the polynomial coefficients and their standard deviations are shown in Table 2. The standard deviations for the parameters were taken from the

Table 1: Model parameters for each term in (3). Source: Coleman et al. (2007).

Parameter	Polynomial
$\log \mu_{max}$	$p_0 + p_1 T$
$\log K_N$	p_2
$\log k_d$	$p_3 + p_4 T + p_5 T^2$
$\log Y_{x/N}$	$p_6 + p_7 N_0$
$\log Y_{e/s}$	p_8
$\log \beta_{max}$	$p_9 + p_{10} T$
$\log K_S$	p_{11}

179 90 % credible region (CR) reported in Coleman et al. (2007) divided by four, i.e. it is
 180 assumed that the distribution is normal and centered around the mean with $\sigma = \text{CR}/4$.

181 2.4. Extended Kalman Filter

182 Extended Kalman filter (EKF), is a recursive algorithm that uses the past mea-
 183 surements and inputs to estimate the current state based on a statistical model. For
 184 estimation, the model given by (1) is augmented with noises on the state and output
 185 equations

$$\begin{aligned} x_{k+1} &= f(x_k, u_k, p, w_k) \\ y_k &= h(x_k) + v_k \end{aligned} \tag{4}$$

186 with $w \in \mathbb{R}^{n_w}$ and $v \in \mathbb{R}^{n_y}$ being random vectors modeled by zero-mean Gaussian noises
 187 with covariance matrices Q and R , respectively. For simplicity of the design, they are
 188 often modeled as non-correlated variables resulting in diagonal matrices containing
 189 the variance for each term. Furthermore, $h(x_k)$ corresponds to the sugar measurement,
 190 which is simply the fourth state plus v_k .

191 Specifically for the fermentation model in (3), the noise w_k is modeled as an addi-
 192 tive term in the parameters as in $\bar{p} + w_k$, where $\bar{p}$ and Q defined by the values in Table 2.
 193 With this design choice, the model used for estimation in this work is as similar as pos-
 194 sible as the one reported in Coleman et al. (2007). It is common practice to include
 195 additive noise in the states when designing Kalman filters. However, they were not

Table 2: Nominal values ($\bar{p}$) and standard deviations ($\bar{\sigma}$) of the parameters. Values in bold are different from Coleman et al. (2007) since they appear to be typos.

	$\bar{p}$	$\bar{\sigma}$
p_0	-3.92	0.11
p_1	7.82 e-2	0.45 e-2
p_2	-4.73	0.14
p_3	-9.81	0.42
p_4	-1.08 e-1	0.40 e-1
p_5	4.78 e-3	0.86 e-3
p_6	3.50	0.098
p_7	-3.61 e-3	0.36 e-3
p_8	-5.98 e-1	0.19 e-1
p_9	-2.98	0.12
p_{10}	7.71 e-2	0.55 e-2
p_{11}	2.33	0.16

196 included in this work since it would require the addition of arbitrary tuning parameters
 197 for the EKF in the form of variance for each noise term.

198 The filter is initialized by two components: the expected mean $\hat{x}_{0|0}$ and the expected
 199 covariance $P_{0|0}$, where the notation $\hat{x}_{k|k-1}$ represents $\hat{x}_k$ given the information at time
 200 $k - 1$. After that, the filter performs two steps at each sampling time: the *prediction* step
 201 and the *update* step. At first, an *a priori* state estimate is calculated by (4). At the same
 202 time, the state noise propagates through the model resulting in an *a priori* covariance
 203 $P_{k|k-1}$. The prediction equations are given by

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, u_{k-1}, \hat{p}) \quad (5a)$$

$$P_{k|k-1} = A_{k-1} P_{k-1|k-1} A_{k-1}^T + G_{k-1} Q G_{k-1}^T \quad (5b)$$

204 where A_{k-1} and G_{k-1} are linearization of the nonlinear model as in

$$A_{k-1} = \left. \frac{df}{dx} \right|_{\hat{x}_{k-1}, u_{k-1}, \bar{p}} \quad G_{k-1} = \left. \frac{df}{dp} \right|_{\hat{x}_{k-1}, u_{k-1}, \bar{p}}$$

205 Notice that G_{k-1} is a derivative to $\bar{p}$ and not w_k since, in this work, the noise appears as

206 $\bar{p} + w_k$. Furthermore, there is no time index in $\bar{p}$ since it is assumed to be constant and
 207 equal to the nominal values.

208 Next, the update step is performed. First, the Kalman gain K_k is calculated using
 209 $P_{k|k-1}$, C_k and R , where C_k is the linearization of $h(\hat{x}_{k|k-1})$. Then, using K_k , the estimates
 210 $\hat{x}_{k|k-1}$ and $P_{k|k-1}$ are updated according to the following equations:

$$K_k = P_{k|k-1} C_k^T (C_k P_{k|k-1} C_k^T + R)^{-1} \quad (6a)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k (y_k - h(\hat{x}_{k|k-1})) \quad (6b)$$

$$P_{k|k} = (I - K_k C_k) P_{k|k-1} \quad (6c)$$

211 Real-world process constraints add additional challenges to the design of state es-
 212 timators for nonlinear systems. For instance, all the states in model (3) must be non-
 213 negative. In the EKF algorithm, there are two steps which can result in invalid values:
 214 the prediction of $\hat{x}_k$ in (5a), and the correction of the estimation in (6b). In this paper, it
 215 is used a simple strategy of clipping negative values to zero (Haseltine and Rawlings,
 216 2005).

217 **Remark 1.** As can be seen in Table 1, the yield coefficient $Y_{X/N}$ depends on the initial
 218 value of N . Hence, for a proper use of the model, the EKF should include online
 219 estimation of initial nitrogen, i.e. $\hat{N}_{0|k}$. On preliminary results, it was found that this
 220 estimation did not significantly affect the overall performance of the filter. Because of
 221 that, in order to simplify the design of the filter, this value is fixed at $\hat{N}_{0|0}$ in this work.

222 2.5. Uncertainty Propagation

223 One of the challenges in nonlinear systems is that there is no easy way to propagate
 224 the covariance matrix P through the model. The EKF attempts to solve this using
 225 the nonlinear model for the estimate of $\hat{x}$ and using a linearized model to calculate
 226 the uncertainty matrix P and the Kalman gain K at each sampling time, as can be
 227 seen in (5b) and (6c). Because of the linearized model for the covariance, EKF might
 228 have poor performance for highly nonlinear systems (Haseltine and Rawlings, 2005).
 229 Nevertheless, the EKF is a straightforward algorithm to generate predictions of the
 230 states $\hat{x}_{t_f|k}$ and their uncertainties $\hat{P}_{t_f|k}$ by sequentially applying (5) until t_f .

Although useful, EKF is not accurate for estimation or uncertainty propagation, specially for severe nonlinearities in the model. Therefore, a more accurate method, such as Monte Carlo (MC) simulations, is required for validation (Akkermans et al., 2018). For the fermentation model, the uncertainty at each sampling time is given by the covariance matrix $P_{k|k}$ for the current state estimate $\hat{x}_{k|k}$, and the covariance matrix Q for the parameter vector p . Using the MC method, a large enough number of scenarios (M) are generated with $\hat{x}_{k|k}$ and p as central points, and $P_{k|k}$ and Q as covariance matrices. Next, each scenario given by a set of x_k^i and p^i , is simulated by ODE solver until t_f resulting in $\mathbf{X}_{t_f} = [x_{t_f}^0, x_{t_f}^1, \dots, x_{t_f}^M]$, while statistical properties can be calculated such as the median, mean $\mu_{t_f} = E[\mathbf{X}_{t_f}]$ or covariance $P_{t_f} = \text{cov}(\mathbf{X}_{t_f}, \mathbf{X}_{t_f})$. Then, this information can be compared to the result of the EKF algorithm in order to assess its accuracy.

A variation of the MC method applied in this work considers only the pair (p, Q) for the scenario generation and assumes knowledge of the current state x_k . This method, referred here as MC0, captures the uncertainty in the prediction caused only by the parameters of the model, removing the effects due to the estimation error of $\hat{x}_k$.

2.6. Proposed Method for Fault Prediction

The proposed method for monitoring and fault prediction, visualized in Figure 2, considers an EKF as an online observer and uses its prediction equations for uncertainty propagation. At each sampling time, the EKF uses the current measurement y_k and the input u_k to update the prediction for $\hat{x}_{k|k}$ and its uncertainty $P_{k|k}$. With this information, the prediction equations for EKF are repeated until t_f resulting in $\hat{x}_{t_f|k}$ and $P_{t_f|k}$. Consequently, at each sampling interval t_s , it is possible to evaluate the probability of a failure occurring at t_f from those values.

In the simulation studies of this work, the MC0 method is applied to the x_k considering only the uncertainty in p . The result from MC0 represents the best achievable performance for the EKF since it removes the effects of estimation error in $\hat{x}_k$. Because of that, MC0 used to validate the prediction provided by the EKF.

Using the information provided by the above methods, it is possible to evaluate, at each t_s , the likelihood of a fault occurring in the future. For the purpose of this

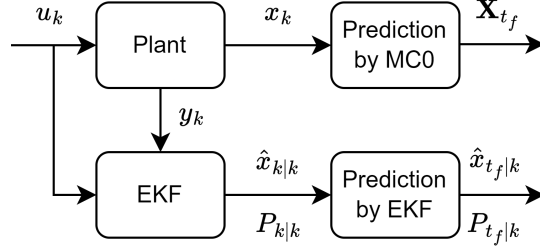


Figure 2: Block diagram summarizing the proposed method for monitoring and fault prediction based on the EKF, and MC0 running in parallel for validation.

paper, which focuses on stuck and sluggish fermentations, a fault is defined as a sugar concentration at t_f above a desired value. For example, a fault could be defined as $S(t_f) \geq 2$ g/L which corresponds to a dry wine, or could be higher for musts that are drained before the end of alcoholic fermentation, which is common for varietal wines (Luna et al., 2021). With this definition, an alarm for a possible fault could be raised at any time when the prediction is above the threshold with a high confidence.

3. Results and Discussion

Simulation studies written in Python are performed to evaluate the performance of the proposed method for wine fermentation. The software SUNDIALS is used as the ODE solver (CVODES) interfaced by CasADi for simulation and automatic differentiation of the sensitivities required by the EKF (Hindmarsh et al., 2005; Andersson et al., 2019). The options for error tolerance were kept on the default values.

The first case study focuses on the performance of the EKF for a specific scenario by considering the nominal parameter values. The second case study is composed of numerous randomly generated MC scenarios where the overall performance of the algorithm is evaluated. Then, the MC scenarios are repeated considering measurement for initial nitrogen. Finally, the results are discussed in further detail in the last section.

3.1. Illustrative Scenario

This simulation study, implements the EKF for fault detection in the nominal case for a white wine scenario with validation of the prediction by the MC0 method.

281 The sampling time used is $t_s = 6$ h with $t_f = 240$ h or 10 days. The desired sugar
 282 concentration at t_f is 10 g/L with a final value higher than that representing a problem-
 283 atic fermentation. The fermentation temperature of 15 °C is constant and known. It is
 284 assumed that **only** measurements for sugar are available, and that they are disturbed by
 285 a zero-mean Gaussian noise with $\sigma_y^2 = 5^2$. The initial values for the states are 0.2 g/L
 286 for biomass, $N_0 = 150$ mg/L, $S_0 = 265$ g/L and zero for ethanol. Both the system and
 287 the estimator have the same nominal parameters given by $\bar{p}_i$ and $\bar{\sigma}_i$ (Table 2). This
 288 scenario results in $S(t_f) = 26.4$ g/L which is a sluggish fermentation.

289 Meanwhile, the initial estimate for the EKF is the same as the process, except for
 290 nitrogen which is significantly different with $\hat{N}_0 = 200$ mg/L. The initial uncertainty P_0
 291 considered in the estimator is given by

$$P_0 = \text{diag}(0.025^2, 0.050^2, 0, 5^2), \quad (7)$$

292 with units in grams per liter. Notice that the standard deviation of 50 mg/L for nitrogen
 293 in the second term is large considering the nominal value, and σ for ethanol is zero,
 294 since it is certain that there is no alcohol at t_0 .

295 Figure 3 shows a histogram of the prediction of the sugar at the end of the fermen-
 296 tation process, $S(t_f)$, using the MC0 method with $M = 2049$. Although MC0 does not
 297 consider the uncertainty in the states, the simulated scenarios result in a wide range for
 298 $S(t_f)$, showing that this system is highly sensitive to the model parameters. A large
 299 portion of the scenarios end successfully with almost no sugar left, but there is a signif-
 300 icant tail which increases the mean and median. The histogram is not bell-shaped and,
 301 because of that, the standard deviation and the mean of the scenarios, estimated by the
 302 EKF, may be biased.

303 Figure 4 shows the estimation of the states by the EKF. In the nominal case, there
 304 is a large error in initial nitrogen at first, but after the **exponential** phase of the fermen-
 305 tation is over, the filter is able to correct the estimated states, resulting in an error close
 306 to zero in steady-state.

307 Figure 5 shows the prediction of final sugar $S(t_f)$ by the EKF and MC0 method
 308 during the 10 days of fermentation. The EKF starts with an optimistic initial estimate
 309 of 200 mg/L for nitrogen and, because of that, the prediction for $\hat{S}(t_f)$ is zero for the

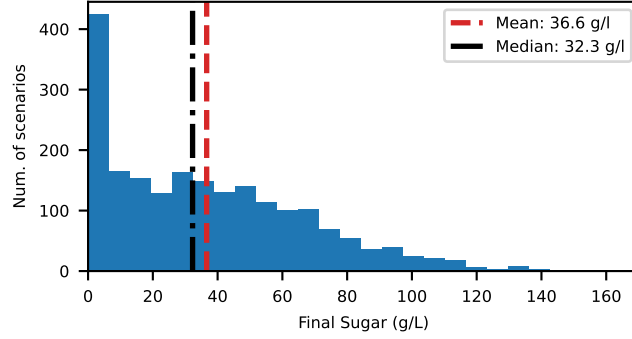


Figure 3: Histogram of $S(t_f)$ using the MC0 method for the nominal model with $n_{MC} = 2049$ and no uncertainty in $x(0)$. For this scenario, $S(t_f) = 26.4$ g/L.

first 60 hours. After the log phase of the fermentation ends and the measurement for sugar starts changing, the filter quickly updates the prediction. This sudden increase in the predicted value for $S(t_f)$ way above the threshold of 10 g/L could be interpreted as an alarm of problematic fermentation. After a few more iterations, the prediction converges to the final value around 26 g/L confirming the problematic fermentation.

Figure 5 also shows two values for the predictions by MC0: the mean and the median. It can be seen that the mean of the MC scenarios is less reliable since the distribution is not normal, resulting in an overprediction of the final sugar. Overall, the MC0 method using the median is more accurate compared to the EKF. However, it still has a considerable error in $S(t_f)$ in the first few days, even though no uncertainty is considered for x_k . This shows that the best achievable performance for the EKF, represented by MC0, can still be quite inaccurate. Despite that, the EKF was still able to generate an alarm in a reasonable time.

3.2. Monte Carlo Scenarios

This section shows Monte Carlo analysis of the proposed EKF for two cases: White Wine (WW) and Red Wine (RW). Each scenario is simulated as described in Section 3.1, but using different initial conditions and model parameter values. Each scenario is defined by the variables described in Table 3 with $t_f = 10$ days for WW and 5 days for RW. The desired sugar at t_f for both cases is 10 g/L. [With this information,](#)

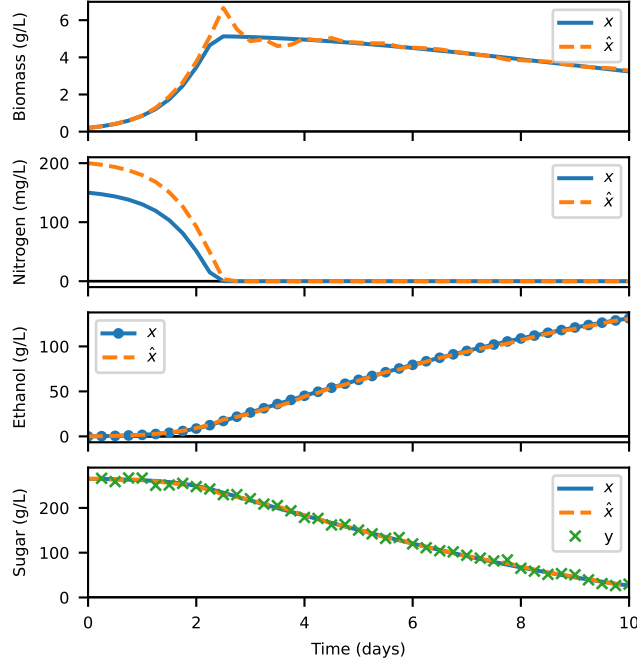


Figure 4: Estimation of the states by EKF for a white wine scenario resulting $S(t_f) = 26.4$ g/L.

329 an alarm is generated at time k when the prediction of final sugar, i. e. $\hat{S}_{t_f|k}$, surpasses
 330 10 g/L. For each scenario, the temperature is constant for the whole batch with 12-18 °C
 331 for whites and 22-28 °C for reds. Scenarios were generated using the Sobol method
 332 from SciPy with $M = 1024$ plus the nominal scenario, totaling 2050 simulations for
 333 both cases. Meanwhile, the EKF is initialized with the same information: nominal N_0
 334 and $\bar{p}$ from Table 3, and P_0 from (7). For 2050 scenarios, 1026 scenarios presented
 335 faults considering the threshold of 10 g/L, with 876 being sluggish fermentation and
 336 150 being stuck.

337 As shown in the case study of Section 3.1, the filter takes a few days to converge
 338 when the initial estimation error is large. This happens around the same time the fer-
 339 mentation leaves the log phase and starts the stationary phase with a faster conversion
 340 of sugar to ethanol. Since EKF depends heavily on sugar measurements, it is expected
 341 that it will only update its model after the process provides meaningful information,
 342 which happens after the fermentation leaves the log phase. This effect results in a large

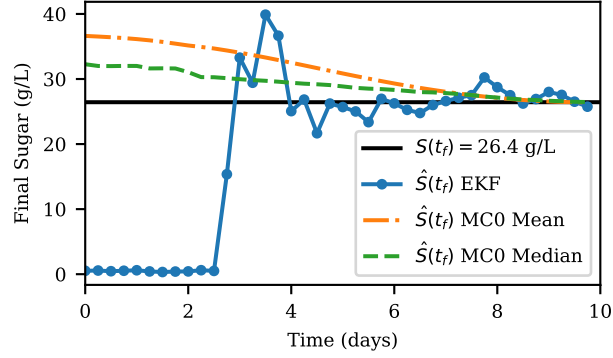


Figure 5: Prediction of the sugar at t_f by EKF and by MC0 (median and mean) compared to the final value of 26.4 g/L.

Table 3: Variables' range for each scenario.

Variable	Nominal	Distribution
N_0	200 mg/L	Uniform [50, 350]
S_0	265 g/L	Uniform [250, 280]
Temp. WW	15 °C	Uniform [12, 18]
Temp. RW	25 °C	Uniform [22, 28]
p	$\bar{p}$	Normal (see Table 2)

343 prediction error for the final sugar on the first days for both RW and WW.

344 Because of that, it is important to ignore the alarms generated before some mini-
345 mum time has passed such as $t_{min} = 24$ h for RW and 48 h for WW. With these criteria,
346 the *first* alarms produced by EKF are summarized in Table 4.

347 The algorithm was successful in 1941 scenarios, where 1010 were scenarios with
348 faults (TP) and 931 with no faults or alarms (TN), resulting in an accuracy of 94.7 %.
349 There were 93 false alarms, which is where the algorithm expects a fault that will not
350 occur, and 16 scenarios where it failed to generate an alarm for a faulty batch. Despite
351 that, the algorithm has a high precision of 91.6 %, meaning that most of the alarms
352 generated were correct. There were 16 scenarios where the EKF failed to generate an
353 alarm for a faulty batch (FN), which can be concerning. In spite of that, the final sugar
354 for these cases was close to the threshold with a median of 11.7 g/L and a maximum of

Table 4: Confusion matrix for the first alarm generated by EKF for both WW and RW cases with 94.7% accuracy, 91.6% precision, 98.4% recall, and 94.9% F1 score.

Total:	Alarm:	No Alarm:
2050	1103	947
Fault:	True Pos.:	False Neg.:
1026	1010	16
No Fault:	False Pos.:	True Neg.:
1024	93	931

355 17.6 g/L.

356 Figure 6 shows histograms of the time of the *first* alarm generated by the EKF for
357 each scenario. For the WW case, most alarms were generated in the first 2–3 days
358 with few outliers. Meanwhile, for the RW case, most alarms were generated between
359 the first and second days with a heavier tail. This histogram shows that it is harder to
360 predict the fermentation for RW probably because of the short duration. However, in
361 both cases, this histogram shows that the algorithm is quite reliable in providing early
362 alarms to the enologist in time for acting on the process.

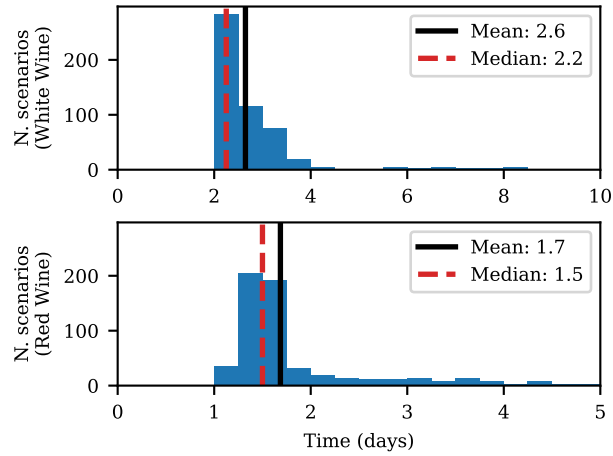


Figure 6: Histogram of time of the first alarm generated by EKF in each scenario, separated by WW (top graph) and RW (bottom graph) cases.

363 For example, in industrial wineries it is common to add nitrogen a second time to

364 ensure a successful process in terms of the productivity and quality. This decision is
 365 carried out typically when the density reaches the range of 1080-1050 kg/m³ that co-
 366 incides with the time range of the first alarm generated (Luna et al., 2021). Since the
 367 number of false positives and false negatives is small, the information is reliable enough
 368 to decide how much nutrient should be added.

369 3.2.1. Measuring Initial Nitrogen

370 This section explores the effects of measuring nitrogen at the beginning of fer-
 371 mentation. All the MC scenarios of the last section are repeated with two differences.
 372 First, the initial variance for nitrogen in P_0 is reduced from 0.050^2 to 0.015^2 which
 373 corresponds to a higher confidence on $\hat{N}_0$. This represents a measurement error for ni-
 374 trogen around ± 30 mg/L which is still significant considering the range for the variable.
 375 Second, at the beginning of each scenario, $\hat{N}_0$ is initialized as N_0 plus a zero-mean
 376 Gaussian noise. This new algorithm is named EKFN0.

377 Figure 7 shows the prediction of $\hat{S}(t_f)$ by EKFN0 for the scenario of Section 3.1
 378 with $\hat{N}_0 = 165$ mg/L compared to EKF with $\hat{N}_0 = 200$ mg/L. When measurement for ini-
 379 tial nitrogen is included, the prediction error for $\hat{S}(t_f)$ decreases significantly. Despite
 380 being far from the real value, the prediction by EKFN0 in the first two days is much
 381 more accurate. In this scenario, an accurate alarm would be available even before the
 382 fermentation start (considering the threshold of 10 g/L).

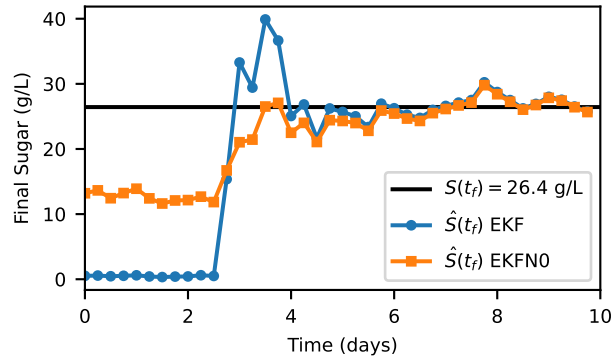


Figure 7: Prediction of the sugar at t_f by EKF and EKFN0 compared to the final value of 26.4 g/L.

383 The performance of EKFN0 in generating alarms is very similar to that of EKF, as

384 reported in the confusion matrix of Table 4. Nearly all scenarios that produced false
385 positives and false negatives with the EKF also yielded the same outcomes with the
386 EKFN0 variant. The main difference was the time of the first alarm for EKFN0. While
387 the EKF gives the results shown in the histogram in Figure 6, EKFN0 generated 87 %
388 of the alarms for WW and 80 % for RW on the first day of fermentation, with a few
389 outliers. As summarized in Table 5, the mean time of the first alarm generated was
390 reduced in 73 % for WW and 65 % for RW. This is a significant improvement which
391 could justify the investment in nitrogen measurement in some cases.

Table 5: Time for the first alarm generated by EKF and EKFN0 for both WW and RW cases.

		First Alarm (days)	
Case		Mean	Median
EKF	WW	2.6	2.2
EKFN0	WW	0.7	0.0
EKF	RW	1.7	1.5
EKFN0	RW	0.6	0.0

392 3.3. Discussion

393 The white wine case study in Section 3.1 showed that the EKF is able to adequately
394 estimate unmeasurable states, such as active biomass. This is crucial since the biomass
395 represents the ability of the yeast to ferment the sugar and avoid faults. Currently, it is
396 impractical to measure active biomass in an industrial setting, so this estimate becomes
397 valuable for the enologist. This information, alongside the knowledge of the expert,
398 can be used to make informed decisions about the fermentation process. For instance,
399 a yearly decline in yeast activity could prompt the enologist to investigate the cause and
400 act accordingly (Malherbe et al., 2007). As shown in the results, the decline in biomass
401 is much more informative than density, which is the most common measurement in
402 wineries.

403 There are two main sources of uncertainty that lead to estimation error. First, there
404 is the problem of uncertainty in the models' parameters that is inherent to the system.
405 The histogram in Figure 3, with the predicted sugar concentration ($S(t_f)$) at the end of

406 fermentation, highlights the sensitivity of the system to the parameters. This aspect can
407 only be improved by developing models that better represent the underlying phenom-
408 ena and by performing experiments to collect data. This can reduce the uncertainty in
409 the model and would improve its prediction accuracy.

410 The second aspect is the use of better algorithms for estimation and prediction.
411 It is known that EKF can show poor performance in some cases, especially for high
412 nonlinearities and constraints (Haseltine and Rawlings, 2005). One reason for that is
413 the use of a first-order method for uncertainty propagation which decreases accuracy
414 and can lead to a poor estimation (Gustafsson and Hendeby, 2012). Although the EKF
415 performed well in this work, it is expected that more advanced methods for uncertainty
416 propagation such as the sigma points or polynomial chaos expansion will lead to better
417 estimation and fault prediction (Akkermans et al., 2018).

418 It is possible to use different methods for estimation and fault prediction. The EKF
419 for estimation could be substituted by the Unscented Kalman Filter for better perfor-
420 mance or by the Moving Horizon Estimation for the use of constraints, although the
421 design process for these methods is more involved. Meanwhile, the uncertainty prop-
422 agation of faults could be carried out online using the MC method, with the downside
423 of increased computational cost. Nevertheless, the process is slow, and the sampling
424 time is large, so this computational overload might not be a practical limitation. These
425 possibilities are worth exploring.

426 Even though the proposed algorithm can be improved, it performed well on the
427 MC scenarios showing robustness under diverse conditions. Providing an accurate
428 prediction of final sugar proved to be a challenge on the first few days of fermentation.
429 The decrease in prediction error around 36 to 48 hours corresponded to the transition
430 from the log phase to the stationary phase, indicating the EKF's reliance on meaningful
431 sugar measurements for effective updates. Still, it was early enough for the process to
432 be acted upon.

433 The histogram shown on Figure 6 with the time of the first alarm across all scenarios
434 provided insight into the reliability of the EKF in providing early alarms. The results
435 indicate that most alarms occur around the average, which typically coincides with
436 the end of the exponential phase. This suggests that, when a fault is detected, it is

likely to be flagged at a meaningful point in the fermentation. However, for shorter fermentations, a higher variability in alarm timing is observed, with more frequent outliers.

The first alarm generated by the algorithm is fairly accurate with 1010 scenarios (91.6 %) being true positive and 93 (8.4 %) false positive, as seen in Table 4. To further increase the reliability of these alarms, one option is to act on the system only after a few alarms are generated in sequence, which decreases the likelihood of a false alarm. Furthermore, the sampling time could be reduced so that more measurements are available. This happens because the EKF has more information available for a better estimation and prediction. Another important aspect is to use the uncertainty $P_{t_f|k}$ alongside $\hat{x}_{t_f|k}$ in the fault detection logic. Hence, an alarm would only be generated when $\hat{S}(t_f)$ is above a threshold and the uncertainty is low enough.

The last case study shows the benefits of measuring initial nitrogen. At least for the model considered, total nitrogen is crucial since it is directly related to peak biomass which, in turn, is necessary for successfully converting all the sugar to ethanol. Measuring nitrogen at the beginning, even with a fairly large error, improves the reliability of the proposed algorithm, although it leads to relevant costs in materials and labor. However, these costs could be reduced by first running the proposed algorithm without measuring N_0 and analyzing the results. If the prediction leads to a successful fermentation with a high degree of certainty, then the measurement could be skipped for that batch.

4. Conclusion

In this study, a monitoring strategy based on EKF for state estimation and early fault detection based on a [dynamic model](#) is proposed in order to provide accurate information during fermentation. This model-based method can support the enologist in detecting problematic batches and designing intervention strategies, while also enabling advanced automatic control which often requires knowledge of all the states of the process. The algorithm was tested in diverse simulation scenarios in order to assess performance and usefulness.

466 Analysis based on MC simulations showed how the process is sensitive to un-
467 certainty on model parameters and initial conditions, especially for nitrogen. It also
468 showed how the estimation is dependent on meaningful measurements for sugar which
469 are not available during the [log](#) phase of the fermentation. Despite these challenges,
470 the main goal of generating accurate early alarms is achieved.

471 There are at least two aspects of the proposed algorithm which can be improved.
472 First, it is expected that more accurate models will result in better estimation, which
473 leads to better fault prediction. Another one is increasing the accuracy uncertainty
474 propagation method for faults. Further analysis could also indicate if effort should be
475 concentrated on designing better estimation and prediction methods, or if it is better
476 to gather more data for developing accurate models. Apart from the computational
477 aspect, it is important to expand the performance analysis of the algorithm since the
478 study was conducted only on simulations. The EKF could be applied to data from
479 industrial batches to check if the conclusions drawn here still hold.

480 The presented findings contribute to the ongoing efforts in leveraging advanced
481 control and monitoring techniques for optimizing wine fermentation processes ([Sablay-
482 rolles, 2009](#)). In an industrial setting, the estimation could be used alongside an auto-
483 matic system for nutrient addition which would autonomously decide how much ni-
484 trogen to add in order to avoid problematic batches. It also enables the use of online
485 model-based controllers such as predictive control. The proposed EKF approach, along
486 with the conducted simulation studies, pave the way for future research in winemaking
487 and analogous processes. Naturally, this includes other alcoholic beverages such as
488 beer, where similar challenges of state estimation and fault detection arise ([Birle et al.,
489 2013](#)). Additionally, estimation and monitoring are increasingly being applied to ad-
490 dress the complexity of other biochemical and biotechnological processes, including
491 bioreactor control, biosensors, and cell culture ([Mowbray et al., 2021](#)). In this context,
492 the proposed framework could be adapted to enhance early fault detection, monitor-
493 ing, and optimization across a broad range of industrial fermentation and biochemical
494 applications, supporting more reliable and sustainable production systems.

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500 Declaration of generative AI and AI-assisted technologies in the writing process

501 During the writing of this work the authors used generative AI tools in order to
502 improve grammar, readability and language of the manuscript. After using these tools,
503 the authors reviewed and edited the content as needed, and take full responsibility for
504 the content of the published article.

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